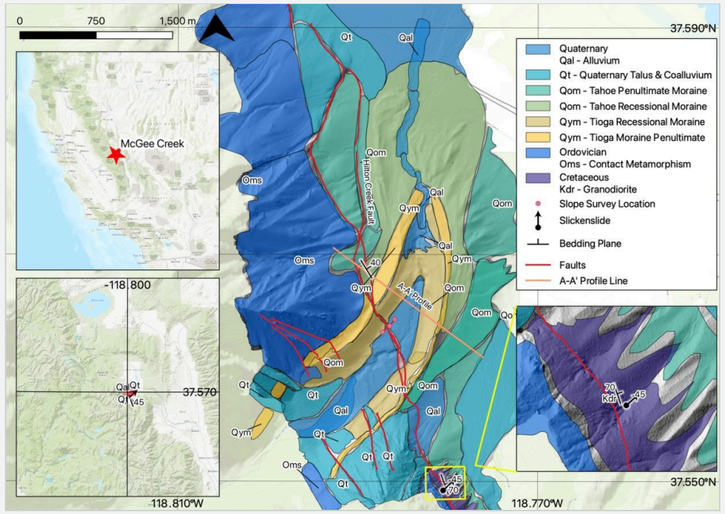
Faulted moraines of McGee Creek

Slip on the Hilton Creek Fault and the relative ages of the Tioga and Tahoe glaciations, eastern Sierra Nevada

Nick Anderson
EPS 118 field report
June 2022



Surficial deposits mapped in the field. Tioga moraines (yellows) sit inside the older Tahoe moraines (greens); red lines are faults. Inset: slope-survey site.

Setting

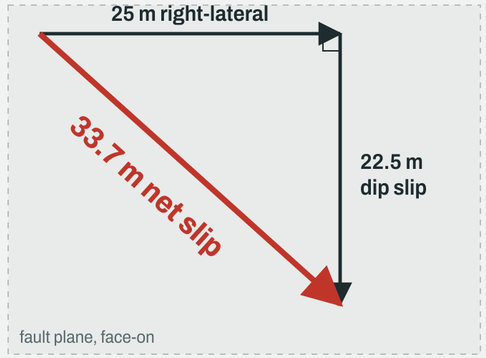
McGee Creek is a U-shaped glacial valley on the eastern Sierra front, just south of Long Valley Caldera. The Hilton Creek Fault, a 22 km range-front fault of the Walker Lane system, cuts moraines of two glaciations: Tahoe (MIS 4–6) and Tioga (MIS 2, the last glacial maximum). Moraines offset by the fault record how far, which way and how fast it has moved.

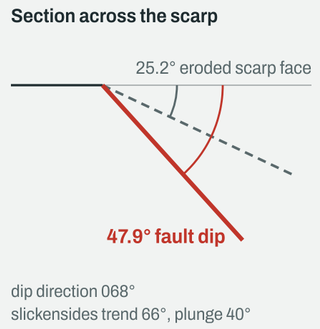
Methods

Faults, slickensides and deposit contacts mapped in FieldMove Clino. A 200 m slope survey across the scarp with sighting level, Brunton, tape and stadia rod, checked against a LiDAR DEM. Clast counts on both moraines for lithology, size, oxidation, burial and xenolith protrusion. Fault-plane fit, offsets and slip computed in QGIS (Qprof) and Python.

33.7 m of oblique slip since the last glacial maximum

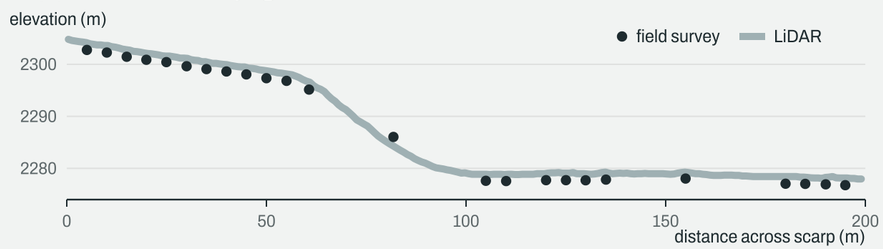
Slip rate through the Tioga moraine: **≈ 1.77 mm/yr** over ~19 kyr





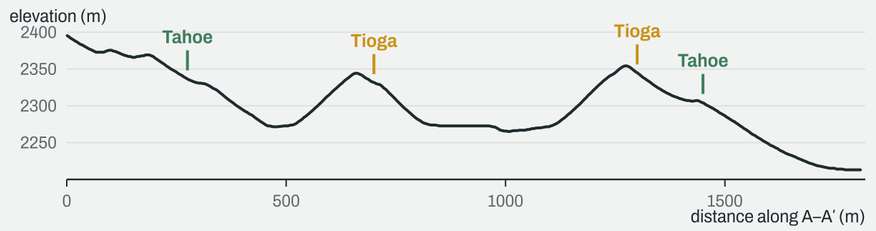
dip direction 068°
slickensides trend 66°, plunge 40°

Scarp profile: field survey against LiDAR

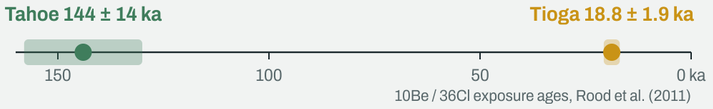


The hand survey and the LiDAR DEM agree along 200 m across the scarp. The eroded scarp face slopes 25.2° today; fitting the fault plane to the DEM gives a dip of 47.9° toward 068°, the angle used for dip slip.

A–A′ topographic profile across both moraines

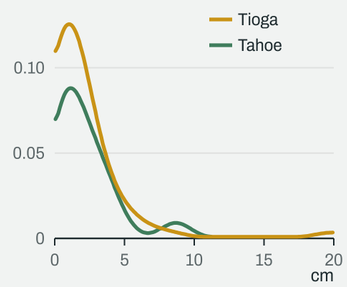


Tioga moraines are younger than the Tahoe moraines, even without exposure dates

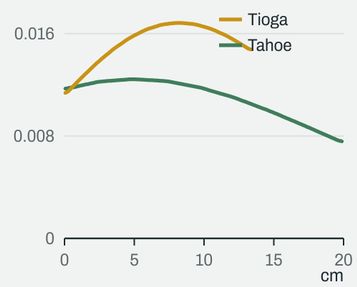


- **Crosscutting.** Sharp Tioga crests cut across the eroded Tahoe moraines on the map.
- **Crest shape.** Tioga crests slope up to ±40°; the rounded Tahoe hills only ±20°.
- **Clast size.** Tioga clasts reach 250 cm² and more. Tahoe clasts average ~6.25 cm² and top out near 175 cm²; most Tahoe metasedimentary clasts are under 2.5 cm long and heavily oxidized.
- **Weathering and burial.** Diorite xenoliths stand out of Tioga granodiorite boulders by less than 5 cm, but more often by over 5 cm on Tahoe boulders. Tioga boulders expose more of their short (z) axis, ~8 cm against ~5 cm, so Tahoe boulders are more deeply buried.

Xenolith protrusion



Exposed boulder z-axis



Crosscutting, crest shape and weathering all put the Tioga moraines after the Tahoe moraines, matching the exposure ages. Slip through the glacial deposits is oblique normal with a right-lateral component (slickensides trend 66°, plunge 40°), at ~1.77 mm/yr. Weak, unconsolidated till along the fault plane could potentially host an earthquake of M6.8.

Berry (1997) Geomorphology 20, 177–195.
Blackwelder (1931) GSA Bulletin 42, 865–922.
Hildreth, Fierstein & Calvert (2018) Quat. Sci. Rev. 198, 242–265.
Rood, Burbank & Finkel (2011) Quat. Sci. Rev. 30, 646–661.
Unruh, Humphrey & Barron (2003) Geology 31, 327–330.
Wahrhaftig et al. (2019) USGS SIM 3414.